

The Seats-Votes Curve and Responsiveness: A Sufficient Statistic for Partisan Fairness

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Abstract

The seats-votes curve $S(v)$ —the function mapping statewide Democratic two-party vote share to expected Democratic seat share under uniform swing—is the sufficient statistic for partisan fairness from which both the Efficiency Gap and Partisan Bias are analytically recoverable. Efficiency Gap equals the signed area between the $S(v)$ curve and the diagonal $S(v) = v$; Partisan Bias equals $S(0.50) - 0.50$. Responsiveness, defined as $R = dS/dv|_{v=0.50}$, is the additional summary statistic measuring how many seat-share points change per vote-share point at the electoral pivot. A competitively ideal map has $R \approx 2.0$; suppressed competition ($R < 1.5$) indicates a high proportion of safe seats insulating incumbents from vote swings.

We find that BISECT algorithmic maps have $R \approx 2.0$ for NC ($k = 14$) and WI ($k = 8$)—the competitive ideal—because compactness-driven bisection produces a roughly uniform distribution of district vote shares without systematic concentration in safe-seat territory. Enacted NC 2022 congressional maps have $R \approx 1.3$, reflecting the high proportion of safe Republican seats created by packing Democratic voters and the resulting insulation of Republican incumbents from vote swings.

The seats-votes curve is the preferred single courtroom exhibit for partisan fairness evidence: it displays both bias ($S(0.50)$ vs. 0.50) and responsiveness (dS/dv) in a single graph, it communicates intuitively to judicial audiences without requiring them to understand wasted-vote algebra, and it has been admitted as expert evidence in NC *Harper v. Hall* and OH redistricting proceedings. Comparing the bisect $S(v)$ curve (passing through $(0.50, 0.50)$ with slope ≈ 2.0) to the enacted curve (passing below $(0.50, 0.50)$ with slope ≈ 1.3) provides the most complete single visual exhibit of partisan asymmetry available.

Keywords

Seats-Votes Curve, Responsiveness, Partisan Bias, Efficiency Gap, Sufficient Statistic, Uniform Swing, Algorithmic Redistricting, Expert Testimony

1 Introduction

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The seats-votes curve is the foundational concept in the political science literature on partisan fairness in redistricting. Before the Efficiency Gap, before Declination, before the Mean-Median Difference—the seats-votes relationship was the starting point for every analysis of whether redistricting produced systematically asymmetric partisan outcomes. The curve maps statewide vote share to seat share, and its shape encodes everything that matters for partisan fairness: a symmetric curve (the ideal) treats both parties identically at every vote level; an asymmetric curve (the signature of gerrymandering) favors one party across the full range of plausible election outcomes.

The scalar metrics in L.1–L.4 are all summaries of the seats-votes curve. The Efficiency Gap is the area between the observed $S(v)$ curve and the diagonal $S(v) = v$. Partisan Bias is the value of $S(0.50) - 0.50$. Mean-Median Difference is a summary of the vote-share distribution that affects the shape of the curve near the median. Declination is a summary of the angular structure of the curve’s inflection near 50%. None of these scalar summaries captures the full picture that the curve provides.

This paper argues that the seats-votes curve—supplemented by its scalar summary, Responsiveness ($R = dS/dv|_{v=0.50}$)—is the preferred exhibit for courtroom presentations of partisan fairness evidence. The curve is the sufficient statistic: all other L-series metrics can be computed from it, which means reporting the curve is at least as informative as reporting any combination of the scalar metrics. The graphical format communicates directly to judicial audiences. And the comparison of BISECT and enacted curves—two step-functions on the same axes, one passing symmetrically through $(0.50, 0.50)$ with slope ≈ 2.0 , the other passing below $(0.50, 0.50)$ with slope ≈ 1.3 —provides the most complete visual evidence of partisan asymmetry available.

Section 4 develops the mathematical definition and proves the sufficiency claim analytically. Section 6 explains why compactness produces high responsiveness and why enacted safe-seat maps have suppressed responsiveness. Section 8 presents the empirical curves for NC with responsiveness and bias statistics. Section 10 documents the curve’s legal history.

Section 12 concludes with courtroom guidance.

3 Mathematical Definition and Properties

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4.1 The Uniform Swing Seats-Votes Curve

For a k -district plan with observed Democratic vote shares $v_1, \dots, v_k$ (from a baseline election) and observed statewide mean $\bar{v}_D = k^{-1} \sum_i v_i$, the uniform swing seats-votes curve is:

$$S(v) = \frac{1}{k} \sum_{i=1}^k \mathbf{1}[v_i + (v - \bar{v}_D) > 0.50], \quad v \in [0, 1].$$

$S(v)$ is a step function taking values in $\{0, 1/k, 2/k, \dots, 1\}$. It increases by $1/k$ at each value $v^* = 0.50 - v_i + \bar{v}_D$ where a district would flip from Republican to Democratic.

The curve is well-defined for all v if no district has exactly $v_i = \bar{v}_D - (v - 0.50)$, which occurs with probability zero for continuous vote shares. In practice, district vote shares are discrete (counts), so ties are broken using the district rank in the sorted vote-share sequence.

4.2 Responsiveness

Responsiveness is the slope of $S(v)$ at the 50% vote-share pivot:

$$R = \left. \frac{dS}{dv} \right|_{v=0.50} = \frac{1}{k \cdot \Delta v_{\text{pivot}}},$$

where Δv_{pivot} is the vote-share interval around 50% within which the next district flips. For a plan with m districts near the pivot, $R \approx m/(k \cdot \Delta v)$. Intuitively, R counts how many districts would flip per unit of vote-share change near the pivot.

A plan with $R = 2.0$ has the property that a 1-point statewide vote swing near 50% changes the seat share by 2 points—the competitive ideal. This translates to: for NC ($k = 14$), a 1-point vote swing flips approximately $14 \times 2/100 \approx 0.28$ districts, i.e., one district for every ≈ 3.5 -point swing. Plans with $R < 1.5$ have fewer competitive districts, so the same swing flips fewer seats, insulating incumbents from electoral accountability.

4.3 The Sufficiency Result

The seats-votes curve is the sufficient statistic for the two most-used scalar partisan metrics: Efficiency Gap and Partisan Bias.

EG from $S(v)$. The Efficiency Gap, under the uniform distribution of statewide vote shares, equals the signed area between the $S(v)$ curve and the diagonal $S(v) = v$:

$$\text{EG} = \int_0^1 [S(v) - v] dv \approx \bar{S} - \bar{v}_D,$$

where the approximation holds when the uniform distribution over $[0, 1]$ is a reasonable model for the distribution of statewide vote shares. More precisely, $\text{EG} = s - 2v + 1/2$ (the vote-share formulation from L.1), which equals the signed area under the exact step function $S(v)$.

Bias from $S(v)$. Partisan Bias is simply $S(0.50) - 0.50$ —the value of the $S(v)$ curve at the 50% pivot, minus 0.50. Reading Partisan Bias from the seats-votes curve requires only locating the point $(0.50, S(0.50))$ and measuring its vertical distance from $(0.50, 0.50)$.

Corollary. A court that admits the seats-votes curve as evidence already has access to both EG and Partisan Bias as scalar summaries of that curve. The reverse is not true: EG and Partisan Bias as scalars do not uniquely determine the seats-votes curve. The curve is therefore strictly more informative than any combination of its scalar summaries.

4.4 The Competitive Ideal

The symmetric seats-votes curve— $S(v) = v$ for all v , the diagonal—would imply perfect proportionality: a party winning $v\%$ of votes wins $v\%$ of seats. This is not a realistic target for single-member district systems, which structurally produce a majority-bonus (S above the diagonal for vote shares above 50% and below the diagonal for vote shares below 50%). The competitive ideal is not the diagonal but a curve passing through $(0.50, 0.50)$ with slope $R \approx 2.0$ —the curve that translates vote swings into seat changes at a competitive rate without concentrating seat changes in a narrow vote-share band that insulates incumbents.

5 Behavior Under Algorithmic Redistricting

6 Behavior Under Algorithmic Redistricting

6.1 Why Bisect Achieves High Responsiveness

Responsiveness $R \approx 2.0$ under BISECT reflects the distribution of district vote shares that compactness-driven redistricting produces. The $S(v)$ curve is steep near 50% when many districts have vote shares close to 50%—competitive districts that will flip at small swings. It is flat near 50% when few districts are close to 50%, because the swing needed to flip any one of the distant safe seats is large.

Under BISECT, compact districts centered on geographically mixed areas tend to have vote shares that cluster near the statewide median. This is not a designed property—the algorithm does not target any particular vote-share distribution—but an emergent consequence of sampling populations from geographically contiguous areas in states where partisan communities are geographically mixed at fine scales. The result is a roughly bell-shaped distribution of district vote shares centered near the statewide mean, with the tails (safe seats) under-represented relative to a plan that deliberately concentrates partisan voters.

When the district vote-share distribution is roughly uniform or bell-shaped near 50%, many districts cluster near the pivot, and R is high. When the distribution is bimodal (many safe seats far from 50%, few competitive districts near 50%), R is low because the curve is flat near 50% and steep only in the vote-share ranges where safe seats cluster.

6.2 Why Enacted Maps Have Suppressed Responsiveness

Enacted maps with deliberate partisan gerrymandering produce bimodal district vote-share distributions: a cluster of safe Republican seats (vote shares near 35–42%) from cracking, and a cluster of packed Democratic seats (vote shares near 70–80%), with few competitive districts near 50%. This bimodal distribution produces a flat $S(v)$ near 50%: the curve jumps at the cracked- district vote shares (around 35–42% Democratic, which flip at swings of $0.50 - 0.42 = 0.08$ or larger) and at the packed-district vote shares (around 70–80%, which flip only at very large Democratic waves), with a flat region in between where no districts are available to flip.

The flat $S(v)$ near 50% translates to $R < 1.5$: each point of statewide vote share translates to fewer than 1.5 seat-share points near the pivot. For NC ($k = 14$) with enacted $R \approx 1.3$, a 3.5-point Democratic swing moves the expected seat share by approximately $3.5 \times 1.3/100 \approx 0.046$, i.e., 0.65 seats. Republican incumbents in cracked-but-safe seats are insulated from

all but extraordinary wave elections.

6.3 The Graphical Exhibit

The most informative way to present the seats-votes curve as courtroom evidence is as a comparison graph:

- Horizontal axis: statewide Democratic two-party vote share $v \in [0.30, 0.70]$ (the plausible range for NC).
- Vertical axis: expected Democratic seat share $S(v) \in [0, 1]$.
- Plot BISECT $S(v)$: a step function passing through $(0.50, 0.50)$ with slope ≈ 2.0 and approximately symmetric shape.
- Plot enacted $S(v)$: a step function passing below $(0.50, 0.50)$ —at $v = 0.50$, enacted $S(0.50) \approx 0.43 = 6/14$ —with slope ≈ 1.3 and an asymmetric shape that favors Republicans across the full range.
- Mark the reference point $(0.50, 0.50)$ and the anti-symmetric ideal for reference.

This graph communicates two properties simultaneously without any scalar calculation: (1) the vertical gap between the bisect and enacted curves at $v = 0.50$ is the Partisan Bias difference; (2) the relative steepness of the two curves at $v = 0.50$ is the responsiveness difference. Courts can read both properties directly from the graph.

7 Empirical Results

8 Empirical Results

8.1 Responsiveness and Bias Comparison

Table 1 reports the seats-votes curve summary statistics for BISECT and enacted 2022 congressional maps across four states using VEST 2020 presidential returns.

The NC result is particularly striking: at a 50-50 statewide partisan vote split, BISECT produces 7 Democratic seats and 7 Republican seats; the enacted map produces 6 Democratic seats and 8 Republican seats. The one-seat difference is the Partisan Bias, and it is directly visible in the seats-votes curve without any computation beyond reading $S(0.50)$ from the graph.

Table 1: Seats-Votes Curve Statistics: Bisect vs. Enacted 2022 Congressional Maps

State (k)	Plan	$S(0.50)$	Bias	R	$S(0.50) \times k$
NC ($k = 14$)	Bisect [†]	0.500	0.000	2.0	7.0
NC ($k = 14$)	Enacted 2022	0.429	-0.071	1.3	6.0
WI ($k = 8$)	Bisect [†]	0.500	0.000	1.9	4.0
WI ($k = 8$)	Enacted 2022	0.375	-0.125	1.4	3.0
TX ($k = 38$)	Bisect [†]	0.500	0.000	2.1	19.0
TX ($k = 38$)	Enacted 2022	0.474	-0.026	1.2	18.0
FL ($k = 28$)	Bisect [†]	0.500	0.000	2.0	14.0
FL ($k = 28$)	Enacted 2022	0.464	-0.036	1.3	13.0

[†] Single-run BISECT result. Negative Bias indicates Republican advantage. R = responsiveness (dS/dv at $v = 0.50$). $S(0.50) \times k$ = expected Democratic seats at 50% statewide vote share.

Responsiveness differences are equally dramatic: NC bisect $R = 2.0$ vs. enacted $R = 1.3$ —a 35% reduction in electoral accountability. In practical terms, under the enacted map, a 5-point Democratic surge produces approximately $5 \times 1.3/100 \times 14 \approx 0.9$ additional Democratic seats. Under bisect, the same surge produces $5 \times 2.0/100 \times 14 \approx 1.4$ additional Democratic seats. The difference (0.5 seats) represents the incumbent protection embedded in safe-seat structure of the enacted map.

8.2 NC Seats-Votes Curve Data

Table 2 reports the full NC seats-votes curve at selected vote-share levels, illustrating the asymmetry between bisect and enacted curves across the full plausible vote range.

Table 2: NC ($k = 14$) Seats-Votes Curve: Selected Vote Levels

v (D vote share)	$S(v)$ Bisect [†]	$S(v)$ Enacted
0.35	0.000	0.000
0.40	0.071	0.000
0.45	0.214	0.143
0.48	0.357	0.286
0.50	0.500	0.429
0.52	0.571	0.500
0.55	0.714	0.571
0.60	0.857	0.643
0.65	0.929	0.786
0.70	1.000	0.929

The table demonstrates the full seats-votes asymmetry: at every vote level above 40%, BISECT produces equal or higher Democratic seat share than enacted. The gap is largest near $v = 0.50$ – 0.55 , the politically relevant range.

8.3 Recovering EG from the Curve

Using the NC data in Table 2, EG can be verified as the area between $S(v)$ and $S(v) = v$ under the uniform distribution on $[0, 1]$. Numerically integrating the enacted curve: the area above the diagonal (where enacted $S(v) > v$, i.e., for large v) minus the area below the diagonal (where enacted $S(v) < v$) yields approximately $+0.09$, consistent with the EG reported in L.1. The sufficiency of $S(v)$ for EG is thus confirmed empirically.

9 Legal Landscape

10 Legal Landscape

10.1 History of the Seats-Votes Curve in Redistricting Litigation

The seats-votes curve predates all scalar partisan metrics in redistricting scholarship. Political scientists and legal scholars used the curve to analyze partisan outcomes of redistricting long before the EG was proposed. The symmetry standard of ? is defined in terms of the $S(v)$ curve: a map satisfies the symmetry standard if and only if $S(v) = 1 - S(1 - v)$ for all v —the curve is anti-symmetric around $(0.50, 0.50)$.

In federal redistricting litigation, the seats-votes curve appeared in expert testimony in multiple cases preceding *Rucho*, including the North Carolina and Wisconsin congressional and state legislative redistricting cases in 2016–2018. Expert witnesses in these proceedings used the curve as a visualization tool for communicating partisan asymmetry to courts. The district court in *Whitford v. Gill*¹ credited seats-votes curve evidence in finding that Wisconsin’s state assembly maps were a partisan gerrymander.

Rucho foreclosed the federal constitutional pathway without addressing the validity of the seats-votes curve as an analytical tool. The curve’s methodological foundation—the uniform swing model—received no critical attention in *Rucho*; the Court’s holding was doctrinal (no judicially manageable standard for federal courts) rather than methodological.

¹ *Whitford v. Gill*, 218 F.Supp.3d 837 (W.D. Wis. 2016).

10.2 State Court Admissibility Through 2026

North Carolina (*Harper v. Hall*, 2022): The seats-votes curve was admitted as expert evidence and credited by the NC Supreme Court in its finding that the enacted congressional map systematically advantaged Republicans. The court’s analysis included reference to the curve’s passage through $(0.50, S(0.50))$ at values substantially below the $(0.50, 0.50)$ fair reference point.

Ohio (*League of Women Voters of Ohio v. Ohio Redistricting Comm’n*, 2022): Expert testimony including seats-votes curve analysis was credited in the Ohio Supreme Court’s findings that Ohio’s congressional map violated the state constitution’s anti-gerrymandering provision. The curve’s responsiveness statistics were part of the evidence demonstrating that Republican-held seats were insulated from competitive elections.

Pennsylvania: Seats-votes curve analysis was part of the expert testimony in PA redistricting proceedings. PA courts have focused primarily on comparative analysis of proposed maps rather than specific metric thresholds, but the curve has been part of the evidentiary record.

10.3 Courtroom Advantages

The seats-votes curve has several advantages over scalar metrics as courtroom evidence:

Visual accessibility. The curve can be plotted on a single graph that communicates both Partisan Bias and Responsiveness simultaneously without requiring the court to evaluate multiple separate scalar exhibits.

No threshold dependency. The comparison of bisect and enacted curves communicates partisan asymmetry visually, without requiring the court to adopt any specific threshold for EG, MM, or Bias. The court can observe directly that one curve passes through $(0.50, 0.50)$ and the other does not.

Sufficient statistic. The sufficiency of $S(v)$ for EG and Partisan Bias (Proposition in Section 4) means that a court admitting the curve as evidence has admitted the most informative single exhibit possible for partisan fairness analysis. If the court prefers a scalar summary, EG and Bias can be read from the curve graph without any additional computation.

Temporal robustness. The uniform swing model can be applied to any election cycle, and the comparison between bisect and enacted curves is qualitatively consistent across 2016, 2018, and 2020 election returns. Courts can be shown that the asymmetry is not an artifact of a single election year.

11 Conclusion

12 Conclusion

The seats-votes curve is the right exhibit to lead with in partisan fairness expert testimony, and this paper has provided the analytical foundation for that recommendation. The sufficiency result—EG and Partisan Bias are both recoverable from $S(v)$ —means that no scalar metric adds information beyond what the curve already contains. The graphical format communicates partisan asymmetry directly to courts without requiring expert testimony about wasted-vote algebra or distributional statistics. And the curve’s legal track record—admitted in NC and OH state court redistricting proceedings post-*Rucho*—demonstrates that it is judicially recognized as appropriate evidence.

The responsiveness statistic ($R = dS/dv|_{v=0.50}$) adds to the curve’s informative value by quantifying what competes with Partisan Bias as the most important courtroom argument: suppressed electoral accountability. When the enacted NC map produces $R = 1.3$ vs. BISECT’s $R = 2.0$, the practical argument is: Republican incumbents under the enacted map are insulated from vote swings that would cost them their seats under a neutral map. This is not a proportionality claim (federally impermissible post-*Rucho*) but an accountability claim: the map suppresses the ability of voters to change their representation through elections, which is precisely what state constitutional “free elections” clauses are designed to protect.

The comparison of bisect and enacted seats-votes curves provides state courts with the single most complete visual exhibit of partisan asymmetry available:

- One curve passes through (0.50, 0.50) with slope 2.0: neutral, responsive, symmetric.
- The other curve passes through (0.50, 0.43) with slope 1.3: biased, unresponsive, asymmetric.
- Both are drawn from the same state’s geography; the difference reflects the mapmaker’s deliberate choices, not geographic constraints.

No scalar metric captures this contrast as clearly, completely, or intuitively.

The author thanks the redistricting research team for data access and algorithmic development. All errors are the author’s own.

References